📘 BE PROJECT REPORT DETAILED INSTRUCTION

◆ Chapter 1 — Introduction (10–12 pages)

Section	What to Write	Pages
1.1 Introduction	Introduce the project domain, explain why this topic is important, describe the area of technology	2
1.2 Background & Motivation	Explain current situation, why project is needed, statistics, challenges, research motivation	2
1.3 Problem Definition	Define exact problem in technical terms, explain gaps in current methods, challenges faced	2
1.4 Proposed Solution	Explain your system, how it solves the problem, high-level working, benefits	2
1.5 Objectives	General objective + 7–10 specific objectives, measurable goals	1
1.6 Significance	Describe benefits to users, society, industry, or academics	1
1.7 Scope	What the system will do, limitations, what it will not cover	1–2

◆ Chapter 2 — Literature Review (15–18 pages)

Section	What to Write	Pages
2.1 Existing Systems	Explain 4–6 existing methods/systems, tools, technologies, pros & cons	4
2.2 Comparative Analysis	Compare existing methods in terms of features, accuracy, cost, speed	3
2.3 Research Paper Review	Summarize at least 8 research papers: author, year, objective, methodology, dataset, results	6

2.4 Research Gap	Clearly identify what is missing in current methods	2
2.5 Justification of Proposed System	Explain why your system is needed, benefits, expected outcomes	2–3

🔹 Chapter 3 — System Analysis & Specifications (15 pages)

Section	**What to Write**	**Pages**
3.1 Introduction	Brief overview of system analysis and need	1
3.2 Problem Statement	Technical definition of problem, scope, constraints	2
3.3 System Overview	Describe modules, workflow, high-level architecture	2
3.4 Functional Requirements	Detailed features list	3
3.5 Non-Functional Requirements	Performance, reliability, security, usability, scalability	3
3.6 Performance Requirements	Expected response time, speed, precision	1
3.7 Design Constraints	Limitations in hardware/software, technology restrictions	1
3.8 System Requirements	Hardware + Software requirements	2

🔹 Chapter 4 — System Planning (10–12 pages)

Section	**What to Write**	**Pages**
4.1 Introduction	Overview of planning and strategy	1
4.2 SDLC Model	Describe model used (Waterfall/Agile), why chosen, phases	3
4.3 Timeline & Milestones	Gantt chart / work breakdown structure	2

4.4 Risk Analysis	List risks, probability, impact, mitigation strategy	2
4.5 Quality Assurance	Testing strategy, code review, validation, standards followed	2
4.6 Monitoring & Control	How project progress will be tracked	1–2

🔹 Chapter 5 — System Design (25–30 pages)

Section	What to Write	Pages
5.1 System Architecture	Diagram, modules interaction, hardware/software interaction	3
5.2 Mathematical Model	Formulas, equations, logic flow	3
5.3 System Flow Diagram	Detailed flow of data & process	3
5.4 ER Diagram	Entities, attributes, relationships	3
5.5 Data Flow Diagrams	Level 0, 1, 2, explanation of data flow	6
5.6 UML Diagrams	Use Case, Class, Sequence, Collaboration, Activity, State chart, Component, Deployment, Package	7–9

🔹 Chapter 6 — System Implementation (18–20 pages)

Section	What to Write	Pages
6.1 Implementation Overview	Step-by-step process of system build	2
6.2 Implemented Architecture	Diagram + explanation of implemented system	3
6.3 Development Environment	OS, IDE, libraries, dependencies	2

6.4 Technologies Used	Languages, frameworks, packages	2
6.5 Module Implementation	Detailed explanation of each module	5–6
6.6 Database Implementation	Tables, schema, queries, sample data	2
6.7 UI Implementation	Screenshots of all interfaces + explanation	2
6.8 Algorithm Implementation	Pseudo code, logic, steps, formula explanation	2–3

◆ Chapter 7 — Testing & Evaluation (10–12 pages)

Section	What to Write	Pages
7.1 Testing Overview	Explain testing purpose	1
7.2 Testing Methods	Unit testing, integration testing, system testing	3
7.3 Test Cases & Results	Table of test cases, inputs, expected outputs, actual outputs	3
7.4 Performance Evaluation	Accuracy, precision, recall, response time analysis	2
7.5 Error Handling	Bugs found, fixes applied	1
7.6 Conclusion of Testing	Summary of test results, performance vs expected	1–2

◆ Chapter 8 — Results & Discussion (6–8 pages)

Section	What to Write	Pages
8.1 Results	Show output screenshots, logs, or analysis	3
8.2 Analysis	Interpret results, compare with expectations	2
8.3 Discussion	Advantages, limitations, challenges faced	1–3

◆ Chapter 9 — Conclusion (3–4 pages)

Section	What to Write	Pages
9.1 Conclusion	Summarize work, achievements, objectives meet	2
9.2 Future Scope	Suggest future improvements, extensions, applications	1–2

◆ Final Sections (5–8 pages)

Section	What to Include	Pages
References	Minimum 50 references, IEEE format, research papers, websites	2–3
Publications	Proof of research papers published	1–2
Certificates	Intercollege Project Competition Participation certificates	3
Plagiarism Report	Turnitin or similar report screenshot	2

Project Report Writing Guidelines

Writing Style

- Use a **mix of paragraphs and bullet points** to make content readable.
- **Paragraphs:**
    - Each paragraph should have 2–5 sentences/ lines.
    - Start with a **topic sentence**, then explain details.
    - Avoid long walls of text — use **subsections and bullet points** for clarity.
- **Clarity & Simplicity:**
    - Write **simple, precise sentences**.
    - Avoid unuse words necessary; explain all **technical terms**.

Diagrams & Illustrations

- Always include **diagrams wherever applicable** (flowchart, UML, ER, screenshots).
- Provide a **caption and 1–2 sentence explanation** for each diagram.
- Use **consistent style** for all diagrams.
- **Reference all figures in text** (e.g., “as shown in Figure 3.1”).

Tables & Lists

- Use **tables for comparisons, test cases, module description, database schema**.
- Number tables sequentially and **refer to them in text**.
- **Lists:**
  - Use **numbered lists** for steps, procedures, or algorithms.
  - Use **bullets** for unordered lists like features or points.

Content Quality

- Include **explanations along with tables, figures, and code snippets**.
- Do not just paste points — **describe each point in 1–2 sentences**.
- Show **analysis and reasoning** wherever applicable (e.g., why a method is chosen).
- For **implementation and results**, include **screenshots with brief explanations**.

Consistency in Language

- Use **consistent terminology** throughout the report.
- Avoid switching terms for the same concept.
- Maintain **consistent tense**: mostly past tense for completed work, present tense for system description.

Proofreading & Review

- Check **spelling, grammar, and sentence clarity**.
- Ensure **figures, tables, and sections are properly labeled and referenced**.
- Avoid redundancy — **do not repeat the same explanation** multiple times.
- Confirm **all objectives are addressed** in the report.

BE Project PPT Content –

Slide No	Slide Title	What to Include / Description	Number of Slides
1	Title Slide	Project Title, Student Names & Roll Numbers, Guide Name, Department & College.	1
2	Introduction	Brief overview of project domain, importance, and motivation (2–3 sentences).	2-3
3	Problem Statement	Define the problem clearly; highlight limitations or challenges in current methods. Optional small diagram.	1
4	Objectives	General objective ,7–10 specific objectives in bullet points.	1
5	Scope	What the system covers and boundaries of the project (5–7 bullet points).	1
6	Significance	Benefits to users, industry, or society (5–7 bullet points).	1
7	Literature Review Overview	Summary of existing systems (2–3 sentences) and comparative analysis.	1
8	Literature Review – Research Gap	Identify missing areas in current methods; justify need for your project.	1
9	Proposed Solution	Brief description of your system; high-level workflow or architecture diagram.	1
10	System Architecture	Diagram; briefly explain modules (1–2 sentences each).	1-2

11	Module 1 – Description	Functionality, key features, brief explanation (paragraph + bullets).	1
12	Module 2 – Description	Functionality, key features, brief explanation (paragraph + bullets).	1
13	Module 3 – Description	Functionality, key features, brief explanation (paragraph + bullets).	1
14	Module 4 – Description	Functionality, key features, brief explanation (paragraph + bullets).	1
15	Module 5 – Description	Functionality, key features, brief explanation (paragraph + bullets).	1
16	Mathematical Model / Algorithms	Key formulas, pseudocode, or flowchart; explain logic briefly.	1-2
17	Database Design	ER diagram, key tables, relationships; brief explanation of schema.	1
18	Data Flow / System Flow	Level 0 / Level 1 flow diagram; explain process in bullets or paragraph.	1
19	User Interface – Overview	Main screens, layout; brief description of functionality.	1
20	User Interface – Screenshots	2–3 screenshots with captions and short explanations.	1–2
21	Implementation Overview	Technology stack (software/hardware), development environment, tools used.	1
22	Testing Strategy	Unit, integration, system testing; bullet points explaining approach.	1
23	Test Cases	Table of key test cases: inputs, expected outputs, actual outputs.	1-2
24	Results	Graphs, charts, screenshots showing outcomes; key findings in bullets.	1–2

25	Discussion	Analyze results; mention advantages and limitations; reasoning behind findings.	1
26	Conclusion	Summary of work, objectives achieved, overall system performance.	1
27	Future Scope	Improvements, extensions, potential applications; 2–3 bullet points.	1
28	References	Key research papers, books, and websites (IEEE format recommended).	1
29	Acknowledgement	Thank guide, institution, collaborators; invitation for questions.	1
30	Q&A / Thank You Slide	Optionally include a visual or project logo; open floor for questions.	1

✅ Notes for Students:

- **Each slides** ensures clarity without overwhelming audience.
- **2–5 bullet points per slide**; **1–2 sentences per point**.
- Include **diagrams, screenshots, and tables** wherever applicable.
- Slides should be **visual and concise**, with **paragraphs only where explanation is necessary**.
- Maintain **uniform font, colour, and layout style** across all slides.
